Apply Basic Mathematics

November/December 2025

SECTION A (40 MARKS)

Attempt ALL the questions in this section.

1. Simply the equation $(y + x) \div 8x - (y - x) \div 4x$. (4 Marks)
2. A boat B starts from a point P on a straight river and moves down the river. An observer O stands at a distance of 300m from P, on the line perpendicular to the river passing through P. After 10min, the observer finds that the angle θ formed by P, himself and the boat, is such that cos θ = 0.7. Determine angle θ and the distance between the observer and the boat at that time. (4 Marks)
3. Mary had 2 bags of flat head nails, each bag containing the same number of flat head nails and five additional spare nails. She put them on a scale and balanced them with 17 single flat head nails. Using the inverse method, solve the equation to give the number of flat head nails in each bag. (4 Marks)
4. Find the vector product of p and q where: $p = 3i - 4j + 2k \ and \ q = 2i + 5j - k$.

   (4 Marks)
5. Solve for x, $log_x 9 + log_{x^2} 3 = 2 \times 5$. (4 Marks)
6. Solve for x in the following equation; $\log(8 - 4x) - 1 = \log(2x - 4)$ (5 Marks)
7. Simplify: $(9s^2 + 3)(s^2 - 4)$ (3 Marks)
8. Using the indices rules, simplify $\frac{6x^{-4} \times 2x^2}{8x^{-3}}$
   (3 Marks)
9. If the mid-point of the line segment joining the points A (3, 4) and B (k, 6) is P (x, y) and x + y – 10 = 0, find the value of k. (4 Marks)
10. Calculate the equation and the gradient of the line joining the points (3, 6) and (2, -5).

    (3 Marks)
11. In a class, 30 students are studying mathematics, 25 are studying carpentry and joinery while 15 are studying both. How many students are in the class? (2 Marks)

SECTION B (60 MARKS)

Attempt Any THREE Questions in This Section

12.

a) Find 3 numbers in geometrical progression whose sum is 28 and whose product is 512.

(8 Marks)

b) If x is real show that $y = \dfrac{x^2+x+1}{x+1}$ is either > 1 or < -3 (5 Marks)

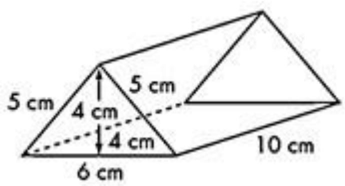
c) Show that $\left(\dfrac{1+\sin x}{1+\cos x}\right)\left(\dfrac{1+\sec x}{1+cosec\, x}\right) = \tan x$ (7 Marks)

13.

a) Solve the simultaneous equation $\begin{aligned} 4x - y &= 1 \\ -2x + 3y &= 12 \end{aligned}$ by matrix method. (7 Marks)

b) Solve $\begin{pmatrix} 4 & 7 & 6 \\ 2 & 3 & 1 \end{pmatrix} \times \begin{pmatrix} 8 \\ 5 \\ 9 \end{pmatrix}$ (5 Marks)

c) Calculate the surface area and volume of the triangular prism below. (8 Marks)



14.

b) Given the currency exchange rates: (4 Marks)

1 Pound = Ksh 145

1 US $ = Ksh 110 Convert:

i. 2180 pounds into Ksh

ii. Ksh 132, 000 to US $.

c) A retailer bought 150 timber boards at Ksh 800 each. He later sold each timber board at

Ksh 1050. Calculate his percentage profit. (5 Marks)

d) The value of a carpentry machine is Ksh 600,000 If it depreciates at a rate of 15% per annum, determine its value after 4 years. (6 Marks)

e) A man invests Ksh 500,000 in savings account that offers simple interest at the rate of 10% per annum. Determine the interest earned after five years. (5 Marks)

15.

a) Find the angle between the vectors a and b given that a = 3i + 4j and b =5i – 12j. Give the answer to 1 decimal place. (9 Marks)

b) A box contains five containers of red paint and three of blue paint. A container is taken out of the box at random. What is the probability that the color of paint is; (3 Marks) i) Red

ii) Blue iii) Green

c) A carpentry workshop contains 6 cutting and 10 sanding machines. If a group of 3 machines is chosen at random from the workshop, find the probability that;

i) 3 sanding machines are selected. (2 Marks) ii) Exactly two sanding machines are selected (2 Marks) iii) At least 1 sanding machine is selected. (2 Marks) iv) Exactly two cutting machines are selected. (2 Marks)